

PHYSICS 101 · RECITATION HOMEWORK

Homework 1: Pressure

Name: _____

Date: _____

Instructions: About 20 minutes. Think in pushes and areas. Part D is a comparison – put the pushes in order and say why. Total: 21 points.

Part A — What pressure is

1. In your own words, what is pressure? (Use the words push and area.) (2 pts)

2. You press down with the same push on a block that covers 1 square, and then on a block that covers 4 squares. Which block gets the higher pressure? How many times higher? (2 pts)

Part B — Match the unit to what it measures

3. Draw a line from each unit to the NICU measurement that uses it. (4 pts)

mmHg
cmH ₂ O
percent (%)
grams (g)

A baby's weight
Blood pressure
How much of the air is oxygen
CPAP pressure

Part C — Torricelli's tube

4. Draw Torricelli's experiment: a dish of mercury with an upside-down tube standing in it, the mercury column inside labeled 760 mm. Next to it, draw a much shorter column and label it with a newborn's top blood pressure number (about 60 mmHg). (2 pts)



Part D — Biggest push to smallest

5. Put these four pushes in order from BIGGEST to SMALLEST, then explain how you decided: (a) the atmosphere at sea level, 760 mmHg; (b) an adult's top blood pressure, 120 mmHg; (c) a newborn's top blood pressure, 70 mmHg; (d) CPAP at 6 cmH₂O (about 4 mmHg). (3 pts)

6. Why do NICU teams use the gentlest ventilator pressure that still works? (1 pt)

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- A. Too much pressure overstretches delicate lungs and can pop an air sac.
- B. Higher pressures make the machine use more electricity.
- C. High pressure pushes the oxygen out through the skin.
- D. Lower pressures make the alarms quieter for other babies.

Part E — Flashback to NICU Tutor

7. A blood pressure is written “64/38.” What is each number called, and what does each one measure? (2 pts)

8. What does CPAP stand for, and what does it do for a preemie's lungs? (2 pts)

Part F — Think like a doctor

9. A preemie on CPAP at 5 cmH₂O is still working hard to breathe — you can see the skin pulling in between the ribs. The team decides to raise the pressure a little, to 6 or 7. Why don't they just turn it way up, to 20, to be safe? (3 pts)

Score: _____ / 21 points

Professor's comment:
